A diagnostic for the linear incidence guide under sparse observation

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Abstract

Stochastic epidemic models of cumulative incidence are used when new infections are recorded in discrete windows and the latent transmission process is not observed. Guiding that latent path inside a particle filter is difficult, because a forward draw of incidence often misses the next count and the weights then degenerate. We give a diagnostic for the linear incidence guide that proposes those latent counts. The guide freezes the hazard over the time remaining until the next observation and then shifts it by a Gaussian approximation of the count. The importance weight keeps the discretised epidemic transition in the numerator. A poor guide changes the weights, whilst the printed target of the ratio stays that transition times the observation probability. We compare the effective sample size of this guide with a myopic proposal on repeated synthetic series, at the published observation spacing and at two coarser spacings fixed before the run. The median effective sample size of the linear guide is smaller at the coarsest spacing than at the published spacing. The myopic comparator does not fall in the same way. We do not replace the guide, and we do not analyse a surveillance series.

Keywords. conditioned hazard, effective sample size, Markov jump process, particle filter, SIR model

1 Introduction

Cumulative incidence is a common observation in epidemic modelling when infection times are hidden and only counts of new cases are recorded in successive windows. Whitaker et al. (2025) take a time-discretised susceptible-exposed-infected-removed model, and its susceptible-infected-removed reduction, as the latent transmission process, and they assimilate noisy inci-

dence with a particle filter. The same construction matters more widely for partially observed jump processes, where a conditioned proposal is used because forward simulation ignores the next observation (Golightly and Wilkinson 2015; Golightly and Sherlock 2019).

The inferential obstacle is an intractable observed-data likelihood. The data are incomplete and discrete, the reporting model is noisy, and the latent incidence between two observation times is a path of counts. A bootstrap filter would draw that path from the unconditioned transition and weight by the observation alone. Whitaker et al. (2025, p. 8) give two reasons not to stop there. Resampling static parameters collapses them, and forward simulation of incidence ignores the next count.

Whitaker et al. (2025) meet the second difficulty with a conditioned hazard. Particles are propagated by replacing the Poisson rate with a hazard derived from a Gaussian approximation of the cumulative incidence between the current time and the next observation, given that observation. The particle is then reweighted by the discretised transition and the observation mass function. The limitations section of that paper states that a further assumption in the derivation is linearity of the latent process over an interval no longer than the observation interval. When the epidemic is sparsely observed, so that cumulative incidence does not evolve linearly, "the conditioned hazard is likely to result in trajectories that are inconsistent with the true conditioned process" (Whitaker et al. 2025, p. 25).

That paragraph does not give a numerical spacing at which the failure begins, and it does not say that the importance weight then targets a different distribution. It names recent work on sampling conditioned jump processes, in particular Golightly and Sherlock (2019) and Corstanje et al. (2023), as relevant and open to adaptation. It does not print either algorithm as a replacement inside the filter.

The contribution here is a diagnostic, in three parts. Firstly, we read the constant-hazard step in the 2025 derivation as the linearity assumption named on that page, and we read the inconsistency as a property of the proposal. Secondly, we report the effective sample size of the incidence weights on the synthetic susceptible-infected-removed mechanism of Whitaker et al. (2025), at the published spacing and at two coarser spacings fixed in advance. Thirdly, we record why the two named guides do not replace the proposal as the filter is written. Unlike Golightly and Sherlock (2019), we do not put a linear-noise bridge for a Gaussian observation of the state into the incidence weight. Unlike Corstanje et al. (2023), we do not replace the generator by a guided process conditioned on an exact end-point.

The illustration is synthetic. Whitaker et al. (2025) also discuss weekly

Ebola counts and weekly COVID-19 counts. Those counts are plotted and are not stored here as a data file, so we do not reanalyse them.

The remainder of this paper is organised as follows. Section 2 compares the guides that were read. Section 3 restates the 2025 filter and the linear incidence guide. Section 4 states what the diagnostic claims. Section 5 reports the simulation. Section 6 records the qualifications and the replacement that remains open.

2 Related work

We compare guides by what they propagate, what they condition on, and whether an importance weight corrects the proposal. Table 1 keeps only rows whose cells were filled from a paper we opened. A paraphrase is not a substitute for that paper.

Equation (9) of Golightly and Wilkinson (2015) is the constant-hazard step that Whitaker et al. (2025) adapt. The observation in the 2015 paper is a Gaussian function of the state, not a binomial or negative binomial count of incidence. Equation (5) of Golightly and Sherlock (2019) restates that hazard. Their Section 3.2 says that normality of the remaining reaction counts, and a hazard that is constant on the whole remaining interval, are often unreasonable. Their Section 6 says that treating the time until the next observation as one discretisation makes the bridges less accurate as the inter-observation time increases. Equation (19) of the same paper is a different proposal: a jump-process bridge under a Gaussian observation of the state, with weight (3). It is not a product of Poisson probabilities on an incidence grid.

Theorem 4.2 of Corstanje et al. (2023) conditions an inhomogeneous Poisson process on an exact integer end-point. The likelihood ratio (16) contains $h(0, x_0)$, the transition probability of the original process. Their Section 4.1 does not treat a binomial observation of an increment on a grid. We return to both constructions in Section 6, where they stay future work.

Lowe et al. (2023) accelerate particle Markov chain Monte Carlo for a chemical kinetic model. On the pages we read, the conditioned hazard is a jump hazard multiplied by a ratio of observation densities, with that density replaced by a linear-noise Gaussian. That account points at Golightly and Sherlock (2019). It is not the incidence proposal in Whitaker et al. (2025). The same pages say that nonlinear behaviour between observations precludes a modified diffusion bridge that pushes particles towards the observations in a linear fashion. They name Durham and Gallant (2002) for that bridge,

and they name Whitaker, Golightly, Boys and Sherlock (2017) for a residual bridge. Neither of those papers was read.

Jovanovski et al. (2024) build data-conditional proposals for approximate Bayesian computation on a diffusion. On the pages we read, the discussion of erratic paths is a conjecture about where their speed-up is largest. It is not a statement that the 2025 incidence hazard fails. Whitaker et al. (2025, pp. 2–3) also name Golightly and Kypraios (2018) beside the bridge. That paper was not read, and we do not use it as a source for the guide.

3 The 2025 filter and the linear guide

Consider the filter of Whitaker et al. (2025). Let X_t denote prevalence and let N_t denote cumulative incidence. The unconditioned hazard is h, and $H = \mathrm{diag}\{h\}$. A scalar observation y_{t_i} is a noisy count of one incidence component over $(t_{i-1}, t_i]$, selected by a vector P. In the susceptible-infected-removed applications, $P' = (1, 0)$. The contact rate enters on the log scale as $\tilde{\beta}_t$, and we use the precision form printed as equation (2) of Whitaker et al. (2025). The static parameter and the particle index are those of their Algorithm 2. We do not restate the gamma updates of their equations (9)–(11). The diagnostic below does not sample them.

Their Algorithm 2 is printed for the negative binomial observation. The synthetic study in their Section 4.1 uses the binomial observation. They do not print a second algorithm box, and we do not invent one. The observation factor in the weight is the mass function of whichever model is in use. In our simulation that factor is the binomial probability mass function.

Algorithm 1. Schematic of Algorithm 2 of Whitaker et al. (2025). The draw from q in step 1c is the object of the diagnostic. It is not replaced.

1. Propagate each particle. Jitter the observation parameter with the kernel printed for the negative binomial model. Draw the contact-rate path from its transition, which the paper calls a blind propagation. Draw the incidence path from the proposal q, which depends on the next observation.

2. Resample with the weight in Equation (1).

3. Update the sufficient statistic for the rate parameters.

4. Draw those rate parameters from their conditional posteriors.

The weight in step 2, for particle k on the window that ends at the next observation, is

$$w^{(k)} \propto \frac{\pi\big(\Delta n^{(k)} \mid \tilde{\beta}^{(k)}, \theta^{(k)}\big)\, \pi\big(y \mid \Delta n^{(k)}, \phi^{(k)}\big)}{q\big(\Delta n^{(k)} \mid \tilde{\beta}^{(k)}, \phi^{(k)}, \theta^{(k)}, y\big)}. \tag{1}$$

The numerator is the time-discretised transition and the observation mass function. It does not contain the conditioned hazard h^*. The denominator is q. We read this ratio as an importance weight for one propagate step. We do not add a convergence theorem for the particle filter.

The proposal q is built in their Section 3.4. At a time τ inside the observation window, equation (12) of Whitaker et al. (2025) approximates the remaining incidence by a Gaussian whose mean and variance are the current hazard times the remaining time. The observation is given a Gaussian approximation, and the mean inside the observation variance is replaced by the mean of that Gaussian. The display after their equation (18) is the conditioned hazard h^*. End-point paths are their forward recursion with h replaced by h^*. The density q is the product of Poisson probabilities with rates $h^*\Delta\tau$.

The printed correction multiplies by $\rho P' H$, a row when the observation is scalar. The cross-covariance block that enters the conditional mean of the remaining incidence is a column. For a diagonal H the row and the column have the same entries. On the susceptible-infected-removed model the shared entries correct the infection component only,

$$h^*_{\text{inf}} = h_{\text{inf}} + \rho h_{\text{inf}} \frac{y - \hat{\mu}}{\rho^2 h_{\text{inf}}(t_i - \tau) + \sigma^2}, \qquad h^*_{\text{rem}} = h_{\text{rem}}, \tag{2}$$

where $\hat{\mu} = \rho\big(n^{\text{past}}_{\text{inf}} + h_{\text{inf}}(t_i - \tau)\big)$ and, for the binomial variance evaluated at $\hat{\mu}$, $\sigma^2 = (1-\rho)\hat{\mu}$. Equation (2) is that scalar form. It matches the entries of the printed display, and it matches equation (9) of Golightly and Wilkinson (2015) in the case where the stoichiometry is the identity, so that the correction is the column HP. It is not a new guide.

The study in Section 5 does not run the jitter, the blind contact-rate draw, the sufficient-statistic update or the conditional parameter draw. The contact-rate path and the prevalence at the left endpoint of the window are taken from the simulated path, and the removal rate and the reporting probability stay at the simulation values. A particle in Algorithm 1 does not have that information. The diagnostic is about h^* given those paths.

4 Diagnostic

Equation (12) of Whitaker et al. (2025) freezes the hazard at the left endpoint of the remaining interval. The mean of the remaining incidence is that frozen hazard times the remaining time. Golightly and Wilkinson (2015, Section 3.1) use the same constant-hazard Gaussian step for remaining reaction counts. Golightly and Sherlock (2019, Section 3.2) say that this step is often unreasonable, and their Section 6 says that accuracy falls as the inter-observation time grows. Whitaker et al. (2025, p. 25) call linearity of the latent process, over an interval no longer than the observation interval, a further assumption in the derivation of the conditioned hazard. Their Section 3.4 does not use the word linear, and it does not print a second condition beyond equation (12) and the display after (18). We do not separate those two passages. The constant-hazard step is the assumption the limitations paragraph is pointing at.

The same paragraph says the conditioned hazard is likely to propose trajectories inconsistent with the true conditioned process, when cumulative incidence does not evolve linearly. The law named on the page is the true conditioned process. The sentence is about the trajectories the hazard proposes. It does not say that the discretised transition in the numerator of Equation (1) is the wrong model, and it does not say that the weight targets a different distribution.

A poor h^* changes q and therefore the weights. On the printed ratio, the importance target of one propagate step stays the discretised transition times the observation mass function. This is a reading of the printed weight. It is not a proof that the particle filter converges.

In that sense, inconsistent is a property of the proposal. No paper we opened states a numerical spacing at which the 2025 guide fails for this epidemic model. Golightly and Sherlock (2019, Section 5.1) report that a linear form fails to describe exponential decay, and that effective sample size falls as the interval length grows, for a death process. We do not copy those figures. They are not results for the epidemic model.

The observable is the effective sample size of the unnormalised incidence weights in one window,

$$\mathrm{ESS}(\tilde{w}_{1:N}) = \frac{\left(\sum_{i=1}^{N} \tilde{w}_i\right)^2}{\sum_{i=1}^{N} \tilde{w}_i^2}, \tag{3}$$

as printed by Golightly and Sherlock (2019, Section 5.2). Zero weights are included. If every weight in a window is zero, Equation (3) is undefined and

the recorded value is 0, as a convention for the table. A draw has weight zero when the correction in Equation (2) is undefined, when a component of the hazard passed to the Poisson draw is negative, or when the updated counts would be negative. Whitaker et al. (2025) do not print a truncation. Golightly and Sherlock (2019, Section 3.2) truncate their equation (5) to keep it positive. We do not copy that truncation onto the 2025 display.

The prediction, fixed before the run, is that the median of this effective sample size across replicate data sets is smaller at the coarsest pre-registered spacing than at the published unit spacing. A myopic proposal, with h^* replaced by h, is a comparator. It is not a second target. The diagnostic will appear as this comparison. It is not a threshold, and it does not say that the filtering distribution changes.

Several claims are outside the paper. We do not replace q by equation (19) of Golightly and Sherlock (2019) or by Theorem 4.2 of Corstanje et al. (2023). We do not drop $h(0, x_0)$ from their ratio (16). We do not present a truncated negative hazard as the 2025 guide. We do not claim a spacing, beyond the three values fixed before the run, at which the guide fails. We do not claim that a poor h^* changes the filtering target. We do not use the death-model table of Golightly and Sherlock (2019) as evidence about the epidemic model. We do not compute a gold-standard posterior.

5 Simulation

The study uses the synthetic susceptible-infected-removed mechanism of Section 4.1 of Whitaker et al. (2025), observed at spacings 1, 2 and 5. Spacing 1 is the published grid. Spacings 2 and 5 are coarser, and they were fixed before any weight was seen. All algorithms were coded in R. No processor is recorded for this run. The data are simulated. There is no gold-standard posterior: the diagnostic does not estimate a posterior distribution of the static parameters.

We drew 20 replicates, with 500 proposals in each window, from seed base 20251015. The seeds are a property of the study script, not of the guide. No fine-grid path was discarded. The run took 46.879 seconds. That time is a record of this run. It is not a comparison with the run times in Whitaker et al. (2025).

For each replicate, each spacing and each guide, the within-replicate summary is the median of the window values of Equation (3). The study summary is the median of those 20 numbers. The prediction uses the linear guide only. The myopic guide is computed on the same simulated paths and

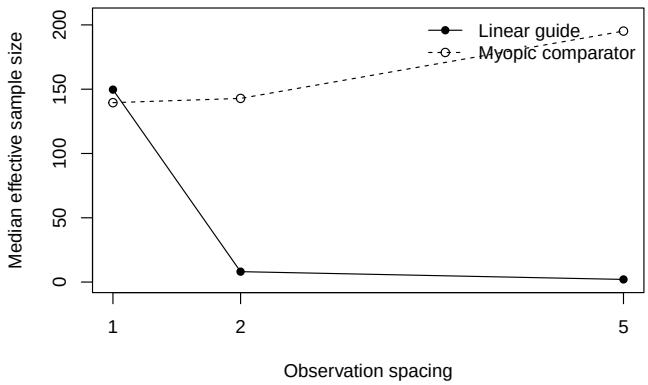


Figure 1: Effective sample size against observation spacing. Filled circles: linear guide, the medians in Table 2. Open circles: myopic comparator, which is not the prediction. The myopic values are those stated in the text.

the same observations, with independent Poisson draws. Table 2 reports the linear guide. Figure 1 plots those three medians and the three myopic medians given below. The myopic series in the figure is a comparator and is not the prediction.

From Table 2, the linear median effective sample size is 149.660679297905 at spacing 1 and 2.05670742297318 at spacing 5. The value at spacing 5 is smaller, so the prediction holds. At spacing 2 the linear median is 8.09166818813341. At every linear row, the median that includes zero weights equals the median computed on the finite weights alone. The fall is present in the finite weights. The median proportion of invalid draws is 0.0329 at spacing 1, 0.009 at spacing 2 and 0 at spacing 5, so the fall at spacing 5 is not produced by the zero-weight rule.

At spacing 1 the myopic median is 139.525563697813. At spacing 2 it is 142.824313796391. At spacing 5 it is 195.181391518181. The myopic median does not fall as the spacing grows.

Two qualifications belong with the comparison. The guide was given the simulated prevalence and the simulated contact-rate path. The result is about h^* given those paths. At spacing 1 the linear median is 149.660679297905 out of 500 draws, so the guide is already uneven on the published grid. The prediction was the comparison across spacings, not a claim that the published grid is unaffected. The comparison does not mea-

sure a pathwise distance to the true conditioned process, and it does not say whether a filtering distribution would change.

6 Discussion

We have given a reading of the linear incidence guide in Whitaker et al. (2025) and a diagnostic for its weights under coarser observation. The constant-hazard step is the linearity assumption named in their limitations section. On the printed importance weight, a poor conditioned hazard changes the proposal and the weights. We have shown, on the synthetic mechanism of their Section 4.1, that the median effective sample size of the linear guide is smaller at spacing 5 than at spacing 1, with the values in Table 2. The myopic comparator does not fall. The guide had the simulated state and the simulated contact-rate path, and even the published spacing already yields a median well below the number of draws.

We expect the diagnostic to be useful when an observation window is lengthened and the question is whether this proposal is the source of uneven weights. We do not expect it to certify a posterior, and we do not treat it as a threshold that divides adequate spacing from inadequate spacing. The three spacings were chosen before the run because they divide the published horizon. They are not estimated change-points.

A replacement of q remains future work. Equation (19) of Golightly and Sherlock (2019) samples a Markov jump process with a linear-noise conditioned hazard, and the weight (3) is the jump-process likelihood ratio under a Gaussian observation of the state. Algorithm 1 divides by the density of Poisson incidence increments on a fixed grid. Equation (19) does not provide that density. Theorem 4.2 of Corstanje et al. (2023) gives the ratio (16) for an inhomogeneous Poisson process conditioned on an exact integer end-point, and (16) contains $h(0, x_0)$. We do not drop that factor, and we do not place either construction in the denominator of Equation (1). Further work needs a proposal whose evaluated density is the density of the same incidence increments that the numerator describes.

The weekly Ebola series and the weekly COVID-19 series in Whitaker et al. (2025) are not analysed. The counts are not stored in this repository, and we do not read them off the published figures. A surveillance example would need those counts in a form that can be reproduced, and a claim that the simulations have not already settled.

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Disclosure statement

The authors report there are no competing interests to declare.

Data availability

All data used in this paper were simulated from the model stated in Section 5. The table and the figure are regenerated by the script named in the supplementary note.

Supplementary material

The supplement for this draft is the base R code that evaluates the linear guide, the study script, the stored tables, and the reproduction script. No surveillance counts are included.

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Table 1: Guides read for this diagnostic. The Whitaker row is the proposal under study. The other rows are not substitutes for it.

Guide	What is propagated	What the proposal conditions on	Weight
Whitaker et al. (2025), display after (18)	Poisson incidence increments on a grid, at rate $h^*\Delta\tau$	The next incidence observation, through a Gaussian approximation	Yes. The ratio uses the discretised transition and the observation mass function over q
Golightly and Wilkinson (2015), equation (9)	A Markov jump process, by exponential waits at the conditioned hazard	A Gaussian observation of the state. The hazard is frozen on the remaining interval	Yes. The complete-data likelihood ratio (10)
Golightly and Sherlock (2019), equation (5)	The same construction as Golightly and Wilkinson (2015), equation (9)	The same Gaussian observation. Section 3.2 says the constant-hazard step is often unreasonable	Yes, through the weight (3), when this hazard is the proposal
Golightly and Sherlock (2019), equation (19)	A Markov jump process, using the linear-noise conditioned hazard without restarting the ordinary differential equations	The same Gaussian observation, through the linear-noise transition density in their equation (6)	Yes. The weight is (3)
Corstanje et al. (2023), Theorem 4.2	An inhomogeneous Poisson process with a homogeneous-Poisson guide	An exact integer end-point	The ratio (16) is printed and contains the transition probability $h(0, x_0)$

Table 2: Linear incidence guide. Median effective sample size across 20 replicates. Finite-weight medians use the draws with a finite log weight. The myopic comparator is not in this table.

Spacing	ESS	ESS, finite weights	Invalid proportion
1	149.660679297905	149.660679297905	0.0329
2	8.09166818813341	8.09166818813341	0.009
5	2.05670742297318	2.05670742297318	0